

Sequences & Mathematical Induction

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5.1 Sequences

5.2 Mathematical Induction I

5.3 Mathematical Induction II



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<http://jarrar-courses.blogspot.com/2014/03/discrete-mathematics-course.html>

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Acknowledgement:

This lecture is based on (but not limited to) to chapter 5 in “Discrete Mathematics with Applications by Susanna S. Epp (3rd Edition)”.

Sequences & Mathematical Induction

5.1 Sequences

In this lecture:



❑ Part 1: **Why we need Sequences (Real-life examples).**

❑ Part 2: Sequence and Patterns

❑ Part 3: Summation: Notation, Expanding & Telescoping

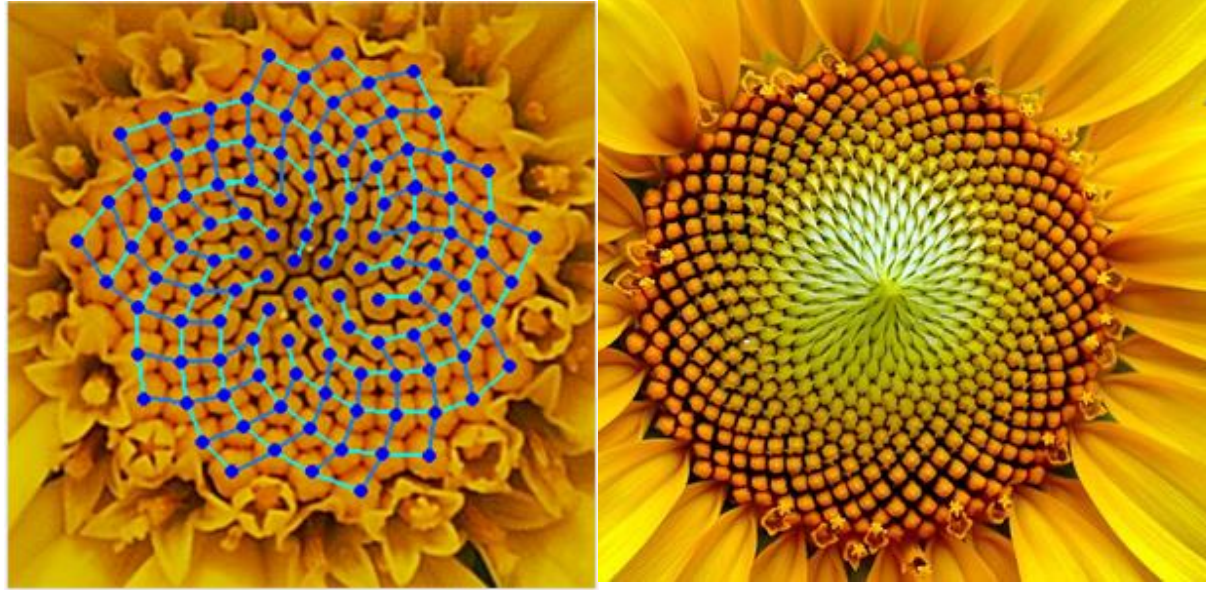
❑ Part 4: Product and Factorial

❑ Part 5: Properties of Summations and Products

❑ Part 6: Sequence in Computer Loops and Dummy Variables

Keywords: Sequences, patterns, Summation, Telescoping, Product, Factorial, Dummy variables,

Motivation



هل يمكن النظر الى علم الرياضيات كعلم اكتشاف انماط في الحياة وتعميم
!! هذه الانماط كنظريات وقوانين؟
!! ما هو المشترك بين الفن وعلم الرياضيات؟

A mathematician, like a painter or poet, is a maker of patterns.

-G. H. Hardy, *A Mathematicians Apology*, 1940

Sequences (المتتاليات)

؟(حتى المستوى الاول :كم عدد أجدادك
؟(حتى المستوى الثاني)
؟(حتى المستوى الثالث)
؟(حتى المستوى الخامس)

Position in the row	1	2	3	4	5	6	7 ...
Number of ancestors	2	4	8	16	32	64	128 ...

؟(K حتى المستوى)

$$A_k = 2^k$$

Train Schedule



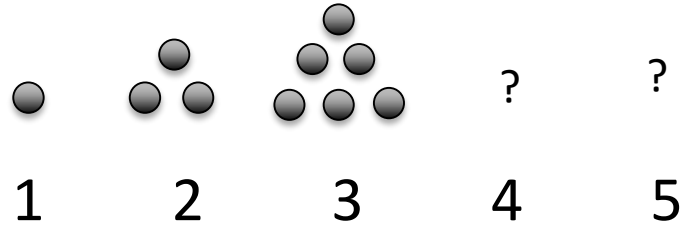
In Nature



<https://www.youtube.com/watch?v=ahXIMUkSXX0>

IQ Tests

Determine the number of points in the 4th and 5th figure



In programming

Any difference between these loops

1. for $i := 1$ to n	2. for $j := 0$ to $n - 1$	3. for $k := 2$ to $n + 1$
print $a[i]$	print $a[j + 1]$	print $a[k - 1]$
next i	next j	next k

$$\sum_{k=1}^n a[k],$$

```
 $s := a[1]$   
for  $k := 2$  to  $n$   
     $s := s + a[k]$   
next  $k$ 
```

```
 $s := 0$   
for  $k := 1$  to  $n$   
     $s := s + a[k]$   
next  $k$ 
```

Sequences & Mathematical Induction

5.1 Sequences

In this lecture:

- ☐ Part 1: Why we need Sequences (Real-life examples).
-  ☐ Part 2: **Sequence and Patterns**
- ☐ Part 3: Summation: Notation, Expanding & Telescoping
- ☐ Part 4: Product and Factorial
- ☐ Part 5: Properties of Summations and Products
- ☐ Part 6: Sequence in Computer Loops and Change of Variables

Keywords: Sequences, patterns, Summation, Telescoping, Product, Factorial, Dummy variables,

Sequences

(المتتاليات)

$$a_m, a_{m+1}, a_{m+2}, \dots, a_n$$

a Sequence is a set of elements written in a row.

Each individual element a_k is called a **term**.

The k in a_k is called a **subscript** or **index**

Finding Terms of Sequences Given by Explicit Formulas

Define sequences $a_1, a_2, a_3, \dots$ and $b_2, b_3, b_4, \dots$ by the following explicit formulas:

$$a_k = \frac{k}{k+1} \text{ for some integers } k \geq 1$$
$$b_i = \frac{i-1}{i} \text{ for some integers } i \geq 2$$

Compute the first five terms of both sequences.

Finding Terms of Sequences Given by Explicit Formulas

Define sequences $a_1, a_2, a_3, \dots$ and $b_2, b_3, b_4, \dots$ by the following explicit formulas:

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$$b_i = \frac{i-1}{i} \text{ for some integers } i \geq 2$$

Compute the first five terms of both sequences.

Solution

ترتيب الحل وترتيب
الافكار مهم جداً لملاحظة
الانماط

$$a_1 = \frac{1}{1+1} = \frac{1}{2}$$

$$a_2 = \frac{2}{2+1} = \frac{2}{3}$$

$$a_3 = \frac{3}{3+1} = \frac{3}{4}$$

$$a_4 = \frac{4}{4+1} = \frac{4}{5}$$

$$a_5 = \frac{5}{5+1} = \frac{5}{6}$$

$$b_2 = \frac{2-1}{2} = \frac{1}{2}$$

$$b_3 = \frac{3-1}{3} = \frac{2}{3}$$

$$b_4 = \frac{4-1}{4} = \frac{3}{4}$$

$$b_5 = \frac{5-1}{5} = \frac{4}{5}$$

$$b_6 = \frac{6-1}{6} = \frac{5}{6}$$

Finding Terms of Sequences Given by Explicit Formulas

Compute the first six terms of the sequence $c_0, c_1, c_2, \dots$

defined as follows: $c_j = (-1)^j$ for all integers $j \geq 0$.

Finding Terms of Sequences Given by Explicit Formulas

Compute the first six terms of the sequence $c_0, c_1, c_2, \dots$ defined as follows: $c_j = (-1)^j$ for all integers $j \geq 0$.

Solution:

$$c_0 = (-1)^0 = 1$$

$$c^1 = (-1)^1 = -1$$

$$c^2 = (-1)^2 = 1$$

$$c^3 = (-1)^3 = -1$$

$$c^4 = (-1)^4 = 1$$

$$c^5 = (-1)^5 = -1$$

Finding an Explicit Formula to Fit Given Initial Terms

Find an explicit formula for a sequence that has the following initial terms:

$$1, -\frac{1}{4}, \frac{1}{9}, -\frac{1}{16}, \frac{1}{25}, -\frac{1}{36}, \dots$$

Finding an Explicit Formula to Fit Given Initial Terms

Find an explicit formula for a sequence that has the following initial terms:

$$1, -\frac{1}{4}, \frac{1}{9}, -\frac{1}{16}, \frac{1}{25}, -\frac{1}{36}, \dots$$

$$a_k = \frac{(-1)^{k+1}}{k^2} \quad \text{for all integers } k \geq 1.$$

OR

$$a_k = \frac{(-1)^k}{(k+1)^2} \quad \text{for all integers } k \geq 0.$$

→ How to prove such formulas of sequences?

Sequences & Mathematical Induction

5.1 Sequences

In this lecture:

- ☐ Part 1: Why we need Sequences (Real-life examples).
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-  ☐ Part 3: **Summation: Notation, Expanding & Telescoping**
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Summation

The diagram illustrates the components of a summation formula. The formula is $\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \dots a_n$. Three orange callout boxes with pointers identify the parts: 'Upper limit' points to the n above the summation symbol; 'Lower limit' points to the $k=m$ below the summation symbol; and 'index' points to the k in the subscript of the summation symbol.

Upper limit

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \dots a_n$$

Lower limit

index

Summation

- Definition

If m and n are integers and $m \leq n$, the symbol $\sum_{k=m}^n a_k$, read the **summation from k equals m to n of a -sub- k** , is the sum of all the terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$. We say that $a_m + a_{m+1} + a_{m+2} + \dots + a_n$ is the **expanded form** of the sum, and we write

$$\sum_{k=m}^n a_k = a_m + a_{m+1} + a_{m+2} + \dots + a_n.$$

We call k the **index** of the summation, m the **lower limit** of the summation, and n the **upper limit** of the summation.

Example

Let $a_1 = -2$, $a_2 = -1$, $a_3 = 0$, $a_4 = 1$, and $a_5 = 2$.
Compute the following:

a. $\sum_{k=1}^5 a_k$

b. $\sum_{k=2}^2 a_k$

c. $\sum_{k=1}^2 a_{2k}$

Example

Let $a_1 = -2$, $a_2 = -1$, $a_3 = 0$, $a_4 = 1$, and $a_5 = 2$.
Compute the following:

a. $\sum_{k=1}^5 a_k$

b. $\sum_{k=2}^2 a_k$

c. $\sum_{k=1}^2 a_{2k}$

Solution:

$$\text{a. } \sum_{k=1}^5 a_k = a_1 + a_2 + a_3 + a_4 + a_5 = (-2) + (-1) + 0 + 1 + 2 = 0$$

$$\text{b. } \sum_{k=2}^2 a_k = a_2 = -1$$

$$\text{c. } \sum_{k=1}^2 a_{2k} = a_{2 \cdot 1} + a_{2 \cdot 2} = a_2 + a_4 = -1 + 1 = 0$$

Example

When the Terms of a Summation are Given by a Formula

Compute the following summation:

$$\sum_{k=1}^5 k^2.$$

Example

When the Terms of a Summation are Given by a Formula

Compute the following summation:

$$\sum_{k=1}^5 k^2.$$

$$\sum_{k=1}^5 k^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = 55.$$

Useful Operations

- Summation to Expanded Form
- Expanded Form to Summation
- Separating Off a Final Term
- Telescoping

➔ These concepts are very important to understand computer loops

Summation to Expanded Form

Write the following summation in expanded form:

$$\sum_{i=0}^n \frac{(-1)^i}{i+1}.$$

Summation to Expanded Form

Write the following summation in expanded form:

$$\sum_{i=0}^n \frac{(-1)^i}{i+1}.$$

Solution

$$\begin{aligned}\sum_{i=0}^n \frac{(-1)^i}{i+1} &= \frac{(-1)^0}{0+1} + \frac{(-1)^1}{1+1} + \frac{(-1)^2}{2+1} + \frac{(-1)^3}{3+1} + \cdots + \frac{(-1)^n}{n+1} \\ &= \frac{1}{1} + \frac{(-1)}{2} + \frac{1}{3} + \frac{(-1)}{4} + \cdots + \frac{(-1)^n}{n+1} \\ &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{(-1)^n}{n+1}\end{aligned}$$

Expanded Form to Summation

Express the following using summation notation:

$$\frac{1}{n} + \frac{2}{n+1} + \frac{3}{n+2} + \cdots + \frac{n+1}{2n}$$

Expanded Form to Summation

Express the following using summation notation:

$$\frac{1}{n} + \frac{2}{n+1} + \frac{3}{n+2} + \cdots + \frac{n+1}{2n}$$

Solution

$$\frac{1}{n} + \frac{2}{n+1} + \frac{3}{n+2} + \cdots + \frac{n+1}{2n} = \sum_{k=0}^n \frac{k+1}{n+k}.$$

Separating Off a Final Term and Adding On a Final Term n

Rewrite $\sum_{i=1}^{n+1} \frac{1}{i^2}$ by separating off the final term.

Write $\sum_{k=0}^n 2^k + 2^{n+1}$ as a single summation.

Separating Off a Final Term and Adding On a Final Term n

Rewrite $\sum_{i=1}^{n+1} \frac{1}{i^2}$ by separating off the final term.

$$\sum_{i=1}^{n+1} \frac{1}{i^2} = \sum_{i=1}^n \frac{1}{i^2} + \frac{1}{(n+1)^2}$$

Write $\sum_{k=0}^n 2^k + 2^{n+1}$ as a single summation.

Separating Off a Final Term and Adding On a Final Term n

Rewrite $\sum_{i=1}^{n+1} \frac{1}{i^2}$ by separating off the final term.

$$\sum_{i=1}^{n+1} \frac{1}{i^2} = \sum_{i=1}^n \frac{1}{i^2} + \frac{1}{(n+1)^2}$$

Write $\sum_{k=0}^n 2^k + 2^{n+1}$ as a single summation.

$$\sum_{k=0}^n 2^k + 2^{n+1} = \sum_{k=0}^{n+1} 2^k$$

Telescoping

A telescoping series is a series whose partial sums eventually only have a fixed number of terms after cancellation [wiki].

متسلسلة لا تحتوي مجاميعها الجزئية في النهاية إلا على عدد ثابت من الحدود بعد الحذف.

Example:
$$\sum_{i=1}^n i - (i+1) = (1-2) + (2-3) + \dots + (n - (n+1))$$
$$= 1 - (n+1)$$
$$= -n$$

This is very useful in programming:

```
S=0
For (i=1;i<=n;i++)
    S= S+ i-(i+1);
```



```
S = -n;
```

Telescoping

A telescoping series is a series whose partial sums eventually only have a fixed number of terms after cancellation [1].

Example:
$$\sum_{k=1}^n \frac{1}{k(k+1)}$$
$$= \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right)$$

```
S=0;  
For (k=1;k<=n;k++)  
    S=S+ 1/k*(k+1);
```

$$= \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{n-1} - \frac{1}{n} \right) + \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$= 1 - \frac{1}{n+1}$$

```
S = 1- (1/(n+1));
```

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- ☐ Part 5: Properties of Summations and Products
- ☐ Part 6: Sequence in Computer Loops and Dummy Variables

Product Notation

- Definition

If m and n are integers and $m \leq n$, the symbol $\prod_{k=m}^n a_k$, read the **product from k equals m to n of a -sub- k** , is the product of all the terms $a_m, a_{m+1}, a_{m+2}, \dots, a_n$.

We write

$$\prod_{k=m}^n a_k = a_m \cdot a_{m+1} \cdot a_{m+2} \cdots a_n.$$

$$\prod_{k=1}^5 k = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 = 120$$

$$\prod_{k=1}^1 \frac{k}{k+1} = \frac{1}{1+1} = \frac{1}{2}$$

Factorial Notation

- Definition

For each positive integer n , the quantity **n factorial** denoted $n!$, is defined to be the product of all the integers from 1 to n :

$$n! = n \cdot (n - 1) \cdots 3 \cdot 2 \cdot 1.$$

Zero factorial, denoted $0!$, is defined to be 1:

$$0! = 1.$$

$$0! = 1$$

$$2! = 2 \cdot 1 = 2$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

$$\begin{aligned} 8! &= 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ &= 40,320 \end{aligned}$$

$$1! = 1$$

$$3! = 3 \cdot 2 \cdot 1 = 6$$

$$5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

$$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5,040$$

$$\begin{aligned} 9! &= 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ &= 362,880 \end{aligned}$$

Factorial Notation

A recursive definition for factorial

$$n! = \begin{cases} 1 & \text{if } n = 0 \\ n \cdot (n-1)! & \text{if } n \geq 1 \end{cases}$$

$$0! = 1$$

$$2! = 2 \cdot 1 = 2$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

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$$\begin{aligned} 9! &= 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ &= 362,880 \end{aligned}$$

Computing with Factorials

$$\frac{8!}{7!} = \frac{8 \cdot \cancel{7!}}{\cancel{7!}} = 8$$

$$\frac{5!}{2! \cdot 3!} = \frac{5 \cdot 4 \cdot \cancel{3!}}{2! \cdot \cancel{3!}} = \frac{5 \cdot 4}{2 \cdot 1} = 10$$

$$\frac{(n+1)!}{n!} = \frac{(n+1) \cdot \cancel{n!}}{\cancel{n!}} = n+1$$

$$\begin{aligned} \frac{n!}{(n-3)!} &= \frac{n \cdot (n-1) \cdot (n-2) \cdot \cancel{(n-3)!}}{\cancel{(n-3)!}} = n \cdot (n-1) \cdot (n-2) \\ &= n^3 - 3n^2 + 2n \end{aligned}$$

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- ☐ Part 6: Sequence in Computer Loops and Dummy Variables

Properties of Summations and Products

Theorem 5.1.1

If $a_m, a_{m+1}, a_{m+2}, \dots$ and $b_m, b_{m+1}, b_{m+2}, \dots$ are sequences of real numbers and c is any real number, then the following equations hold for any integer $n \geq m$:

$$1. \sum_{k=m}^n a_k + \sum_{k=m}^n b_k = \sum_{k=m}^n (a_k + b_k)$$

$$2. c \cdot \sum_{k=m}^n a_k = \sum_{k=m}^n c \cdot a_k \quad \text{generalized distributive law}$$

$$3. \left(\prod_{k=m}^n a_k \right) \cdot \left(\prod_{k=m}^n b_k \right) = \prod_{k=m}^n (a_k \cdot b_k).$$

➔ Remember to apply these in programing Loops

Example

Let $a_k = k + 1$ and $b_k = k - 1$ for all integers k . Write each of the following expressions as a single summation or product:

$$\sum_{k=m}^n a_k + 2 \cdot \sum_{k=m}^n b_k$$

$$\prod_{k=m}^n a_k \cdot \prod_{k=m}^n b_k$$

Example

Let $a_k = k + 1$ and $b_k = k - 1$ for all integers k . Write each of the following expressions as a single summation or product:

$$\sum_{k=m}^n a_k + 2 \cdot \sum_{k=m}^n b_k$$

$$\begin{aligned} \sum_{k=m}^n a_k + 2 \cdot \sum_{k=m}^n b_k &= \sum_{k=m}^n (k + 1) + 2 \cdot \sum_{k=m}^n (k - 1) \\ &= \sum_{k=m}^n (k + 1) + \sum_{k=m}^n 2 \cdot (k - 1) \\ &= \sum_{k=m}^n ((k + 1) + 2 \cdot (k - 1)) \\ &= \sum_{k=m}^n (3k - 1) \end{aligned}$$

$$\prod_{k=m}^n a_k \cdot \prod_{k=m}^n b_k$$

$$\begin{aligned} \left(\prod_{k=m}^n a_k \right) \cdot \left(\prod_{k=m}^n b_k \right) &= \left(\prod_{k=m}^n (k + 1) \right) \cdot \left(\prod_{k=m}^n (k - 1) \right) \\ &= \prod_{k=m}^n (k + 1) \cdot (k - 1) \\ &= \prod_{k=m}^n (k^2 - 1) \end{aligned}$$

Sequences & Mathematical Induction

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-  ☐ Part 6: **Sequence in Computer Loops & Change of Variables**

Keywords: Sequences, patterns, Summation, Telescoping, Product, Factorial, Dummy variables,

Change of Variable

Observe: $\sum_{k=1}^3 k^2 = 1^2 + 2^2 + 3^2$ $\sum_{i=1}^3 i^2 = 1^2 + 2^2 + 3^2.$

Hence: $\sum_{k=1}^3 k^2 = \sum_{i=1}^3 i^2.$

Also Observe: $\sum_{j=2}^4 (j-1)^2 = (2-1)^2 + (3-1)^2 + (4-1)^2$
 $= 1^2 + 2^2 + 3^2$
 $= \sum_{k=1}^3 k^2.$

Replaced Index by any other symbol (called a **dummy variable**).

Programing Loops

Any difference between these loops

- | | | |
|--------------------------------------|--|--|
| 1. for $i := 1$ to n | 2. for $j := 0$ to $n - 1$ | 3. for $k := 2$ to $n + 1$ |
| print $a[i]$ | print $a[j + 1]$ | print $a[k - 1]$ |
| next i | next j | next k |

$$\sum_{k=1}^n a[k],$$

```
 $s := a[1]$   
for  $k := 2$  to  $n$   
     $s := s + a[k]$   
next  $k$ 
```

```
 $s := 0$   
for  $k := 1$  to  $n$   
     $s := s + a[k]$   
next  $k$ 
```

Change Variables

Transform the following summation by making the specified change of variable.

$$\sum_{k=0}^6 \frac{1}{k+1} \quad \text{Change variable } j = k+1$$

```
For (k=0; k≤6; k++)  
Sum = Sum + 1/(k+1)
```

$$\sum_{j=1}^7 \frac{1}{j} = \sum_{k=1}^7 \frac{1}{k}.$$

$$\sum_{k=0}^6 \frac{1}{k+1} = \sum_{k=1}^7 \frac{1}{k}$$

```
For (k=1; k≤7; k++)  
Sum = Sum + 1/(k)
```

Change Variables

Transform the following summation by making the specified change of variable.

$$\sum_{k=1}^{n+1} \frac{k}{n+k}$$

```
For (k=1; k<=n+1; k++)  
Sum = Sum + k/(n+k)
```

Change of variable: $j = k - 1$

$$\sum_{j=0}^n \frac{j+1}{n+(j+1)} = \sum_{k=0}^n \frac{k+1}{n+(k+1)}$$

$$\sum_{k=1}^{n+1} \frac{k}{n+k} = \sum_{k=0}^n \frac{k+1}{n+(k+1)}$$

```
For (k=0; k<=n; k++)  
Sum = Sum + (k+1)/(n+k+1)
```

Programing Loops

All questions in the exams will be loops

Thus, I suggest:

Convert all previous examples into loops and play
with them